

Niloy Banerjee

Kolkata, India

Data Science | Data Analytics | Consumer Analytics | Market Intelligence

Data Analytics and Consumer Insights professional with 12+ years of experience in FMCG analytics, market intelligence, business strategy, and advanced market research across ITC, Nielsen, Ipsos, and Kantar. Extensive experience in leveraging primary research, syndicated data, consumer panels, retail analytics, and advanced statistical techniques to generate actionable business insights and support strategic decision-making.

Strong expertise in consumer behaviour analytics, segmentation, brand health tracking, communication testing, product & pack testing, shopper analytics, pricing & promotion analysis, marketing effectiveness, and market mix modelling. Proven track record of translating complex business problems into data-driven solutions that drive growth, improve marketing ROI, and strengthen consumer understanding.

Skilled in Python, SQL, Databricks, Pandas, NumPy, Scikit-Learn, machine learning, predictive analytics, statistical modelling, regression, classification, clustering (K-Means), feature engineering, model evaluation, forecasting, A/B testing, and time-series analytics including LSTM-based approaches. Hands-on experience with data visualization, dashboard development, analytics automation, and business intelligence solutions.

Additionally, developed multiple Data Science, Machine Learning, NLP, Computer Vision, and AI-powered applications including forecasting models, intelligent analytics assistants, sentiment analysis systems, RAG-based knowledge systems, and multi-agent workflows using LLMs, LangChain, LangGraph, vector databases, and modern AI frameworks. Currently expanding expertise in cloud analytics platforms (Azure, AWS), data lakes, Snowflake, scalable data architectures, and enterprise AI solutions.

Core Skills

Market Research & Consumer Analytics

Brand health tracking, communication testing, product & pack testing, segmentation, usage & attitude (U&A), shopper analytics (observation & eye tracking), satisfaction & driver analysis, retail audit (Nielsen RMS), household panel analytics (Kantar HHP).

Data Analytics & Data Science

Exploratory data analysis (EDA), descriptive statistics, hypothesis testing, predictive analytics, regression, classification, clustering (K-Means), feature engineering, model evaluation, forecasting, market mix modelling, consumer behaviour analytics.

Programming, Visualization & Platforms

Python, SQL, Databricks, Snowflake (Learning), Pandas, NumPy, Scikit-Learn, Jupyter Notebook, Streamlit, Tableau, Matplotlib, Seaborn, SPSS, Advanced Excel, Git, GitHub.

Machine Learning & AI

Machine learning, neural networks, deep learning fundamentals, LSTM, NLP, transformer models, LLM applications, RAG, vector databases, AI agents, LangChain, LangGraph, Data Lakes, automated analytics systems.

Domain Expertise

FMCG analytics, consumer analytics, market intelligence, marketing analytics, pricing & promotion analytics, business intelligence, cross-functional strategy support.

Technical Stack

Python | SQL | Databricks | Snowflake (Learning) | Git | GitHub | Pandas | NumPy | Scikit-Learn | Tableau | Streamlit | SPSS | Excel

Machine Learning | Predictive Analytics | XGBoost | Regression | Classification | Clustering | Feature Engineering | Forecasting | ARIMA | Prophet | LSTM | A/B Testing | Market Mix Modelling

NLP | Transformers | RAG | Vector Databases | LangChain | LangGraph | AI Agents

Azure (Learning) | AWS (Learning) | Data Lakes | Analytics Automation

Data Science & AI Projects

Consumer Segmentation Engine

Developed a customer segmentation framework using K-Means clustering, behavioural analytics, and demographic variables to identify high-value consumer segments and support targeted marketing strategies.

Marketing Mix Analytics & Optimization

Built statistical models to evaluate the impact of advertising, pricing, promotions, and distribution variables on sales performance, enabling data-driven marketing investment decisions.

Retail Demand Forecasting

Developed forecasting models using ARIMA, Prophet, and LSTM techniques to predict product demand, sales trends, and inventory requirements across multiple scenarios.

AI-Powered Analytics Assistant

Designed and developed multiple AI-powered analytics assistants using Python, LLMs, LangChain, LangGraph, vector databases, and agentic AI frameworks to automate business analytics workflows. Key capabilities included:

- Natural language query processing
- Automated insight generation
- SQL query generation and execution
- Report generation and summarization
- Multi-agent orchestration
- Business analytics automation

AI Research & Knowledge Assistant

Built RAG-based knowledge systems using vector databases, embeddings, and LLMs to enable intelligent document search, question answering, and knowledge retrieval.

NLP-Based Sentiment Analysis System

Developed transformer-based NLP pipelines to analyze sentiment and extract insights from financial, consumer, and market-related text datasets.

Consumer Analytics Dashboard

Built interactive analytics dashboards and automated reporting solutions using Python, Tableau, and visualization libraries to monitor business performance, consumer trends, and market KPIs.

Computer Vision & Autonomous Navigation Research

Developed computer vision prototypes combining object detection, depth estimation, and motion analysis techniques for autonomous navigation and intelligent decision-support applications.

Professional Experience

ITC Limited

Manager – Shopper, Consumer & Trade Research

Dec 2018 – Present

Lead consumer analytics and market intelligence initiatives for ITC's Foods and Personal Care portfolio, leveraging large-scale consumer, retail, and syndicated datasets to drive strategic business decisions.

Key Contributions

- Performed advanced consumer, market, and retail analytics using Nielsen RMS, Kantar HHP, and custom research datasets to identify growth opportunities and market trends.
- Designed and executed segmentation models, shopper analytics, U&A studies, and satisfaction frameworks to understand consumer behaviour and support targeted business strategies.
- Applied statistical analysis, driver modelling, and predictive analytics techniques to evaluate pricing effectiveness, promotional performance, and key growth drivers.
- Delivered data-driven recommendations supporting category strategy, marketing investment optimization, innovation pipelines, and portfolio planning.
- Developed automation solutions, analytics dashboards, and reporting workflows using Python, Tableau, and AI-powered tools to improve efficiency and decision-making.
- Collaborated with cross-functional teams across marketing, product, sales, and strategy functions to translate business problems into analytical solutions.
- Developed analytics automation, forecasting, and AI-assisted insight generation solutions using Python, SQL, machine learning, and NLP techniques.

Nielsen India Pvt. Ltd.

Research Manager

Mar 2015 – Dec 2018

Led large-scale consumer analytics and market intelligence programs for leading FMCG and retail organizations.

Key Contributions

- Managed consumer segmentation, brand equity, pricing, and shopper analytics studies across multiple product categories.
- Analyzed large-scale retail audit, household panel, and custom research datasets to generate actionable business insights.

- Applied quantitative techniques including statistical modelling, driver analysis, segmentation, and trend forecasting to support strategic decision-making.
- Delivered executive-level insights and recommendations to client leadership teams through data-driven storytelling and business analytics.
- Managed stakeholder relationships and coordinated multi-functional teams to ensure successful project delivery.

TNS India Pvt. Ltd. (Kantar)

Senior Market Research Executive

Jan 2014 – Mar 2015

- Conducted consumer analytics, brand equity, and market tracking studies across FMCG categories.
- Developed brand health tracking frameworks and performance monitoring systems for clients and competitors.
- Analyzed consumer and market data to generate actionable insights supporting marketing and growth strategies.

Ipsos Market Research

Market Research Executive

Jul 2012 – Jan 2014

- Conducted brand communication and consumer analytics studies for leading brands.
- Analyzed research datasets to identify market trends, consumer perception, and brand performance drivers.
- Delivered analytical insights and strategic recommendations through structured reporting and presentations.

JSW Steel

Assistant Manager – Coke Oven Greenfield Project

Aug 2007 – Jul 2010

- Project planning, vendor coordination, and execution activities in a large industrial project.

Education

Advanced Programme in Data Science (Executive)

Indian Institute of Management Calcutta – 2025-26

PGDM – Marketing

Welingkar Institute of Management Development & Research - 2012

B.Tech – Ceramic Technology

Government College of Engineering & Ceramic Technology - 2007

Certifications

Data Science Bootcamp – Certification

365 Data Science – 2025

AI Engineering

365 Data Science – 2025-26

Python Programming

Completed certification courses from:

- Udemy
- PythonX
- United Latino Students Association

Awards & Recognition

ITC Achiever & Applause Awards – FY 2021–22, 22-23, 25-26

Nielsen “Simple Excellent” Gold and Silver Awards for project management and research delivery.

TNS Spirit Award for client management and research excellence.

Additional Achievements

- Designed and developed **RC and autonomous drones and aircraft for scientific applications**.
- Research work on computer vision and advance analysis from frame-by-frame data analysis
- Researching on Automation and Robotics using multi sensor data analysis combined with AI Agents, LLM, Python and Gen – AI
- Multiple awards for Drone Building and Drone Racing at national level from across states.
- Trained Tabla player and award-winning painter.
- Organized state-level science exhibition during academic years.